Impact of micro-mesoporous structures on TENGs

"How do mixed micro-mesoporous structures influence TENG performance" *Generated by LeapSpace*

Mixed **micro-mesoporous structures generally improve TENG performance** by combining the strengths of both pore scales: **micropores** add charge-trapping and interfacial-polarization sites, while **mesopores** increase accessible surface area, contact area, and charge-transport pathways; the best performance usually comes from a **balanced hierarchical pore network**, not from maximizing either pore size alone. [46] [86] [87] [88] [89] [96]

How the two pore scales contribute

Structural feature	Main TENG effect	Net influence
Micropores	Provide **high-density confinement sites** for trapped charge and strengthen interfacial polarization, which helps retain triboelectric charges longer. [87] [90] [91] [92]	**Strong for charge retention**, but can be limited if too restrictive or overly blocking. [87] [90]
Mesopores	Increase **effective surface area**, improve contact electrification, and create **shorter diffusion/transport pathways** for charge and carriers. [46] [86] [88] [93] [94]	**Usually the bigger output driver** because they more directly raise usable interfacial area. [46]
Micro + meso together	Combine **trapping + transport + surface-area gain**, giving higher charge density and more stable output than either scale alone. [92] [95] [96] [97]	**Best overall when well connected and not over-porous**. [87] [97]

What this means in practice

- Hierarchical porous TENG layers can raise output because more interfaces and void surfaces support **charge accumulation and retention**. [46] [92] [97] [98]
- In porous dielectric layers, output improvement is not just from more contact area; **capacitance changes** also matter and can strongly shift open-circuit voltage. [96] [96] [99]
- The same hierarchy can help both **charge generation** and **charge stability**: micro-level confinement traps charges, while mesopores keep the structure accessible enough for repeated contact-separation cycling. [88] [90] [97]
- If the pore network becomes too open, too thick, or too defective, performance can fall because charge transfer becomes less efficient and structural continuity is reduced. [87] [93] [100]

Bottom line

Mixed micro-mesoporous structures are beneficial for TENGs because they fuse the trapping advantage of microporosity with the area/transport advantage of mesoporosity. The most defensible design target is a **hierarchical, well-connected pore network** that maximizes interfacial polarization and charge storage while preserving enough continuity for efficient charge transfer and stable cycling. [46] [86] [87] [90] [97]

Confidence level: **High** — multiple retrieved studies and reviews converge on the same hierarchy-of-benefits pattern for porous TENG materials, with the main nuance being the need to avoid excessive porosity or structural blockage.

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